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Discrete optical unit on a substrate of an integrated photonics chip

Abstract

An integrated photonics device that emits light out towards a measured sample value is disclosed. The device can include a discrete optical unit that attaches to a supporting layer. The discrete optical unit can include mirror(s), optics, detector array(s), and traces. The supporting layer can include one or more cavities having facet walls. Light emitter(s) can emit light that propagate through waveguide(s). The emitted light can exit the waveguide(s) (via termination point(s)), enter the one or more cavities at the facet walls, and be received by receiving facets of the discrete optical unit. The mirror(s) of the discrete optical unit can redirect the received light towards collimating optics, which can direct the light out of the device through the system interface. The discrete optical unit can be formed separately from the supporting layer or bonded to the supporting layer after the mirror, optics, detector arrays, and traces are formed.

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Background/Summary

CROSS REFERENCE OF RELATED APPLICATIONS (1) This application is a continuation of U.S. patent application Ser. No. 17/254,810, filed Dec. 21, 2020, which a national stage application under 35 U.S.C. § 371 of PCT/US2019/038599, filed Jun. 21, 2019, which claims the benefit under 35 U.S.C. § 119(e) of U.S. Provisional Patent Application No. 62/689,018, filed on Jun. 22, 2018, the contents of which are incorporated herein by reference as if fully disclosed herein.

FIELD

(1) This disclosure relates generally to an integrated photonics device configured for measuring one or more properties of a sample volume. More specifically, the integrated photonics device can

include a discrete optical unit formed on a substrate.

BACKGROUND

(2) Optical sensing systems can be used for a number of applications, one can be to measure one or more properties of a sample volume. The optical sensing system can include an integrated photonics device comprising a plurality of optical components such as light sources and detectors. In some instances, the integrated photonics device can be arranged such that light emitted from the light sources can propagate through a waveguide in one plane (e.g., x-y plane) and can be redirected to another plane (e.g., y-z plane) by a mirror (e.g., outcoupler). The mirror can have an angled wall to redirect incident light, where the angled wall can be formed by etching one or more layers, such as a waveguide. In some instances, the one or more layers may have one or more crystallographic planes, so etching may lead to a limited number of possibilities for angles.

(3) The etching may also limit the height of the mirror, which may limit the area of its reflecting surface. The optical sensing system can have a light beam that exits a termination point of a waveguide, which may diverge starting at the termination point. The divergence of the light beam can cause a change in its size, shape, and/or symmetry relative to the termination point. This divergence may lead to an incident light beam having a size that is greater than and/or a centroid that is offset from the reflecting surface of the mirror. When the reflecting surface of the mirror is smaller than the size of the incident light beam and/or its centroid is offset from the centroid of the light beam, a portion of the incident light may be lost and/or may include one or more undesirable properties (e.g., spectral ripples, high efficiency loss, stray light issues).

(4) The amount of beam divergence may differ depending on the wavelength(s) of the emitted light. Since some wavelengths may have a greater divergence angle than others, the beam properties (e.g., light loss and control) may be spectrally-dependent. A discrete optical unit suitable for an integrated photonics device that can account for the effects from beam divergence and fully capture the light beam may be desired.

SUMMARY

(5) Described herein is an integrated photonics device used for emitting light out of a device towards a measured sample value, where a portion of the emitted light can be returned and measured for determining one or more properties of the measured sample volume. The integrated photonics device can include a discrete optical unit that attaches to a supporting layer. The discrete optical unit can include one or more of mirror(s), optics, detector array(s), and traces. The supporting layer can include one or more cavities having facet walls. One or more light emitters can emit light that propagates through one or more waveguides. The emitted light can exit the waveguide(s) (via termination point(s)), enter the one or more cavities at the facet walls, and be received by receiving facets of the discrete optical unit. The mirror(s) of the discrete optical unit can redirect the received light towards collimating optics. The collimating optics can direct the light out of the device through the system interface.

(6) The discrete optical unit can be formed separately from the supporting layer and bonded to the supporting layer after one or more of the mirror, optics, detector arrays, and traces are formed in its substrate. In this manner, the reflecting surface(s) of the mirror(s) may be sized and located to reduce loss of light and its control. In some examples, a fill material can be deposited between the facet wall of the cavity and the discrete optical unit to change the amount of divergence of the emitted light beam. Examples of the disclosure further include forming one or more overhangs in the discrete optical unit to reduce light loss.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A illustrates a block diagram of an exemplary optical sensing system according to

examples of the disclosure.

(2) FIG. 1B illustrates an exemplary process flow for measuring sample properties according to examples of the disclosure.

(3) FIG. 2A illustrates a cross-sectional view of an exemplary portion of an integrated photonics device according to examples of the disclosure.

(4) FIG. 2B illustrates an exemplary process for the operation of a portion of an integrated photonics device according to examples of the disclosure.

(5) FIG. 2C illustrates a cross-sectional view of an exemplary portion of an integrated photonics device including an overmold layer according to examples of the disclosure.

(6) FIG. 3A illustrates a top view of a portion of an integrated photonics device including multiple cavities according to examples of the disclosure.

(7) FIG. 3B illustrates a planar view of a portion of a discrete optical unit including multiple mirrors and attachment portions according to examples of the disclosure.

(8) FIG. 3C illustrates a top view of an exemplary discrete optical unit including detector arrays according to examples of the disclosure.

(9) FIG. 4A illustrates a cross-sectional view of an exemplary portion of an integrated photonics device including a fill material according to examples of the disclosure.

(10) FIG. 4B illustrates a cross-sectional view of an exemplary portion of an integrated photonics device having a discrete optical unit with an angled receiving facet according to examples of the disclosure.

(11) FIG. 4C illustrates a cross-sectional view of an exemplary portion of an integrated photonics device having a supporting layer with an angled facet wall according to examples of the disclosure.

(12) FIG. 4D illustrates a cross-sectional view of an exemplary portion of an integrated photonics device having a supporting layer with a ledge according to examples of the disclosure.

(13) FIG. 4E illustrates a cross-sectional view of an exemplary portion of an integrated photonics device having a supporting layer with a pedestal according to examples of the disclosure.

(14) FIG. 4F illustrates a cross-sectional view of an exemplary portion of an integrated photonics device having a discrete optical unit with an attachment portion having an angled wall according to examples of the disclosure.

(15) FIG. 4G illustrates an exemplary discrete optical unit that includes a prism according to examples of the disclosure.

(16) FIG. 5A illustrates an exemplary process for fabricating an exemplary discrete optical unit according to examples of the disclosure.

(17) FIG. 5B illustrates an exemplary process for forming the supporting layer and attaching the discrete optical unit to the supporting layer according to examples of the disclosure.

(18) FIG. 6A illustrates a top view of an exemplary integrated photonics device including a plurality of waveguides and a common mirror according to examples of the disclosure.

(19) FIG. 6B illustrates a top view of an exemplary integrated photonics device including a plurality of waveguides and individual mirrors according to examples of the disclosure.

(20) FIG. 6C illustrates a top view of an exemplary integrated photonics device including a plurality of waveguides and a plurality of mirrors formed on a single discrete optical unit according to examples of the disclosure.

(21) FIG. 6D illustrates a top view of an exemplary supporting layer including a plurality of cavities having staggered facet walls according to examples of the disclosure.

DETAILED DESCRIPTION

(22) Optical sensing systems can be used for a number of applications, one can be to measure one or more properties of a sample volume. The optical sensing system can include an integrated photonics device comprising a plurality of optical components such as light sources and detectors. In some instances, the integrated photonics device can be arranged such that light emitted from the light sources can propagate through a waveguide in one plane (e.g., x-y plane) and can be

redirected to another plane (e.g., y-z plane) by a mirror (e.g., outcoupler). The mirror can have an angled wall to redirect incident light, where the angled wall can be formed by etching one or more layers, such as a waveguide. In some instances, the one or more layers may have one or more crystallographic planes, so etching may lead to a limited number of possibilities for angles.

(23) The etching may also limit the height of the mirror, which may limit the area of its reflecting surface. The optical sensing system can have a light beam that exits a termination point of a waveguide, which may diverge starting at the termination point. The divergence of the light beam can cause a change in its size, shape, and/or symmetry relative to the termination point. This divergence may lead to an incident light beam having a size that is greater than and/or a centroid that is offset from the reflecting surface of the mirror. When the reflecting surface of the mirror is smaller than the size of the incident light beam and/or its centroid is offset from the centroid of the light beam, a portion of the incident light may be lost and/or may include one or more undesirable properties (e.g., spectral ripples).

(24) The amount of beam divergence may differ depending on the wavelength(s) of the emitted light. Since some wavelengths may have a greater divergence angle than others, the beam properties (e.g., light loss and control) may be spectrally-dependent. A discrete optical unit suitable for an integrated photonics device that can account for the effects from beam divergence and fully capture the light beam may be desired.

(25) In the following description of examples, reference is made to the accompanying drawings in which it is shown by way of illustration specific examples that can be practiced. It is to be understood that other examples can be used and structural changes can be made without departing from the scope of the various examples.

(26) Various techniques and process flow steps will be described in detail with reference to examples as illustrated in the accompanying drawings. In the following description, numerous specific details are set forth in order to provide a thorough understanding of one or more aspects and/or features described or referenced herein. It will be apparent, however, to one skilled in the art, that one or more aspects and/or features described or referenced herein may be practiced without some or all of these specific details. In other instances, well-known process steps and/or structures have not been described in detail in order to not obscure some of the aspects and/or features described or referenced herein.

(27) Further, although process steps or method steps can be described in a sequential order, such processes and methods can be configured to work in any suitable order. In other words, any sequence or order of steps that can be described in the disclosure does not, in and of itself, indicate a requirement that the steps be performed in that order. Further, some steps may be performed simultaneously despite being described or implied as occurring non-simultaneously (e.g., because one step is described after the other step). Moreover, the illustration of a process by its description in a drawing does not imply that the illustrated process is exclusive of other variations and modification thereto, does not imply that the illustrated process or any of its steps are necessary to one or more of the examples, and does not imply that the illustrated process is preferred.

(28) Described herein is an integrated photonics device used for emitting light out of a device towards a measured sample value, where a portion of the emitted light can be returned and measured for determining one or more properties of the measured sample volume. The integrated photonics device can include a discrete optical unit that attaches to a supporting layer. The discrete optical unit can include one or more of mirror(s), optics, detector array(s), and traces. The supporting layer can include one or more cavities having facet walls. One or more light emitters can emit light that propagates through one or more waveguides. The emitted light can exit the waveguide(s) (via termination point(s)), enter the one or more cavities at the facet walls, and be received by receiving facets of the discrete optical unit. The mirror(s) of the discrete optical unit can redirect the received light towards collimating optics. The collimating optics can direct the light out of the device through the system interface.

(29) The discrete optical unit can be formed separately from the supporting layer and bonded to the supporting layer after one or more of the mirror, optics, detector arrays, and traces are formed in its substrate. In this manner, the reflecting surface(s) of the mirror(s) may be sized and located to reduce loss of light and its control. In some examples, a fill material can be deposited between the facet wall of the cavity and the discrete optical unit to change the amount of divergence of the emitted light beam. Examples of the disclosure further include forming one or more overhangs in the discrete optical unit to reduce light loss.

(30) Representative applications of methods and apparatus according to the present disclosure are described in this section. These examples are being provided solely to add context and aid in the understanding of the described examples. It will thus be apparent to one skilled in the art that the described examples may be practiced without some or all of the specific details. Other applications are possible, such that the following examples should not be taken as limiting.

(31) An overview of the components included in an exemplary integrated photonics device and operation thereof are now described, with detailed descriptions provided below. FIG. 1A illustrates a block diagram of an exemplary optical sensing system according to examples of the disclosure. FIG. 1B illustrates an exemplary process flow for measuring sample properties according to examples of the disclosure. The system **100** can include an interface **180** (e.g., a system interface), an optical unit **190**, a light emitter **107**, a detector **130**, and a controller **140**. The system **100** can include an integrated photonics device, and the interface **180** can include an external surface of the device, which can accommodate light transmission through it, among other things. The interface **180** can include an emission region **182**, a reference **108** (optional), and a detection region **156**. In some examples, the emission region **182** can include one or more components (e.g., an aperture layer) configured to limit the optical path lengths and/or angles of light entering the system **100**. By limiting the optical path lengths and/or angles of light, the light incident on, or exiting from, a measured sample volume **120** can also be limited. Optical unit **190** can include an absorber or light blocker **192**, optics **191** (e.g., lenses), optics **194** (e.g., a negative microlens), and light collection optics **116** (e.g., a positive microlens). In using the system, the measured sample volume **120** can be located near, close to, or touching at least a portion (e.g., interface **180**) of the system **100**. The light emitter **107** can be coupled to the controller **140**. The controller **140** can send a signal (e.g., current or voltage waveform) to control the light emitter **107**, which can emit light (step **153** of process **150**). The light emitter **107** can include a plurality of waveguides, in some examples. The light emitter **107** can emit light towards the emission region **182** (step **155** of process **151**).

(32) The emission region **182** can be configured to allow light to exit the system **100** towards the measured sample volume **120**. Depending on the nature of the measured sample volume **120**, light can penetrate a certain depth into the measured sample volume **120** to reach one or more scattering sites and can return (e.g., reflect and/or scatter back) towards the system **100**. The return light can enter back into the system **100** at the detection region **156** (step **159** of process **151**). The return light that enters back into the system can be collected by light collection optics **116**, which can direct, collimate, focus, and/or magnify the return light (step **161** of process **151**). The return light can be directed towards the detector **130** (e.g., a detector array). The detector **130** can detect the return light and send an electrical signal indicative of the amount of detected light to the controller **140** (step **163** of process **151**).

(33) The light emitter **107** can optionally emit light towards the reference **108** (step **165** of process **151**). The reference **108** can redirect light towards optics **194** (step **167** of process **151**). The reference **108** can include, but is not limited to, a mirror, a filter, and/or a sample with known optical properties. Optics **194** can direct, collimate, focus, and/or magnify light towards the detector **130** (step **169** of process **151**). The detector **130** can measure light reflected from the reference **108** and can generate an electrical signal indicative of this reflected light (step **171** of process **151**). The controller can be configured to receive at least two electrical signals from the detector **130**. In some instances, one electrical signal can be indicative of return light from the

measured sample volume **120**, and another electrical signal can be indicative of light reflected from the reference **108**. The different electrical signals can be a time-multiplexed signal, for example. The electrical signal at a given instance in time can be based on whether light is sent to the measured sample volume or the reference. In other instances, the two or more electrical signals can be received from different detector pixels simultaneously and may include different light information. The controller **140** (or another processor) can determine the properties of the sample from the electrical signals (step **173** of process **1.51**).

(34) In some examples, when the system is measuring the properties of the sample and the reference, light emitted from the light emitter **107** can reflect off a surface of the sample back into the system **100**. Light reflected off the interior walls or components can be referred to as the interface reflected light **184**. In some examples, the interface reflected light **184** could be light emitted from the light emitter **107** that has not reflected off the measured sample volume **120** or the reference **108** and can be due to light scattering within the system **100**. Since the interface reflected light **184** can be unwanted, the absorber or light blocker **192** can prevent the interface reflected light **184** from being collected by optics **194** and light collection optics **116**. In this manner, the system can prevent the interface reflected light **184** from being measured by the detector **130**.

(35) Arrangement of a Portion of an Exemplary Integrated Photonics Device

(36) A description of a portion of an exemplary integrated photonics device is now provided. FIG. 2A illustrates a cross-sectional view of an exemplary portion of an integrated photonics device according to examples of the disclosure. In using the device **200**, the measured sample volume (e.g., measured sample volume **120** illustrated in FIG. 1A) can be located close to the system interface **280**. The device can include one or more windows (not shown) located at or proximate to the system interface **280**. The window(s) can include one or more transparent and thermally conductive materials such as sapphire, silicon, or a combination thereof.

(37) The measured sample volume can include one or more locations, which can include one or more scattering sites associated with scattering event(s). The device **200** can be configured to reconstruct the optical paths in the measured sample volume. For example, the device **200** can be configured to reconstruct angles and locations of light received at the detection regions (e.g., detection regions **156** illustrated in FIG. 1A) to another place (e.g., a plane located closer to the detector **130** illustrated in FIG. 1A). One or more layers of optics including optics **291** (discussed below) can be included in the device **200** and used for reconstructing the optical paths.

(38) The device **200** can include multiple components, where the multiple components can be formed on, attached to, touching, or coupled to (e.g., indirectly associated with) a wafer, the wafer can include a supporting layer **242**. The supporting layer **242** can include any type of material such as silicon. At least some of the multiple components can include optical components. Exemplary optical components can include a light emitter (e.g., light emitter **107** illustrated in FIG. 1A), a detector array (e.g., detector **130** illustrated in FIG. 1A), receiving optics (e.g., optics **194** illustrated in FIG. 1A), collecting optics (e.g., light collection optics **116** illustrated in FIG. 1A), and one or more discrete optical units **243**. Other components (not shown) can include optical traces, multiplexers, amplifiers, and the like.

(39) The supporting layer **242** and/or one or more layers **210** can include one or more cavities **221** at one or more locations along the wafer. The cavity **221** can include one or more components within its recess (discussed below). The walls of the cavity **221**, which can include a facet wall **222**, can define the recess of the cavity. Exemplary components can include, but are not limited to, at least a portion of the discrete optical unit **243**, adhesive layer **212**, fill material, and the like. The cavity **221** can be formed by removing (e.g., etching) material from the supporting layer **242** and/or one or more layers **210**. In some examples, the cavity **221** can be formed by selectively depositing one or more of the layers **210** at a limited number of locations along the wafer of the device **200**. In some instances, the facet wall **222** of the cavity can include an anti-reflection coating **223**.

(40) The adhesive layer **212** can be any type of material that allows the discrete optical unit **243** to

bond to the supporting layer **242**. Exemplary materials can include, but are not limited to, glue (e.g., thermal epoxy) and solder (e.g., gold, tin, etc.). In instances where solder is used, the adhesive layer **212** can include a plated material, such as nickel and/or gold, deposited on the supporting layer **242**. Additionally or alternatively, the adhesive layer **212** can also include a patterned solder, such as gold tin, deposited on the discrete optical unit **243**.

(41) The device **200** can include a system interface **280**. The system interface **280** can include one or more emission regions (e.g., emission region **182** illustrated in FIG. **1A**) and one or more detection regions (e.g., detection region **156** illustrated in FIG. **1A**). The emission region(s) can be configured to allow light **241** emitted by the light emitter, redirected by the mirror **209** (included in the discrete optical unit **243**), and collected by optics **291** to exit the device **200** at the system interface **280**. The detection region(s) (e.g., detection region **156** illustrated in FIG. **1A**) can be configured to allow return light to enter the device **200** to be redirected by the collecting optics (e.g., light collection optics **116** illustrated in FIG. **1A**) and detected by the detector array (e.g., detector **130** illustrated in FIG. **1A**). In some examples, certain detector pixels included in the detector array can be associated with different optical path lengths to determine (e.g., estimate) the optical properties (e.g., absorbance) of the measured sample volume.

(42) The device **200** can include one or more light emitters (e.g., light emitter **107** illustrated in FIG. **1A**). A light emitter can be configured to emit light **241**. The light emitter can include any type of light source and one or more waveguides capable of generating and transmitting light **241** to the discrete optical unit **243**, which can be located at least partially in the cavity **221**. In some instances, the light emitter can include a single light source. In other instances, the light emitter can include a plurality of discrete light sources. A light source can include, but is not limited to, a lamp, laser, light-emitting diode (LED), organic light-emitting diode (OLED), electroluminescent (EL) source, quantum dot (QD) light emitter, super-luminescent diode, super-continuum source, fiber-based source, or a combination of one or more of these sources. In some examples, the light emitter can be capable of emitting a single wavelength of light. In some examples, the light emitter can be capable of emitting a plurality of wavelengths of light. In some examples, the light emitter can include any tunable source capable of generating a short-wave infrared (SWIR) signature. In some examples, a light emitter can include a III-V material, such as Indium Phosphide (InP), Gallium Antimonide (GaSb), Gallium Arsenide Antimonide (GaAsSb), Aluminum Arsenide (AlAs), Aluminum Gallium Arsenide (AlGaAs), Aluminum Indium Arsenide (AlInAs), Indium Gallium Phosphide (InGaP), Indium Gallium Arsenide (InGaAs), Indium Arsenide Antimonide (InAsSb), Indium Phosphide Antimonide (InPSb), Indium Arsenide Phosphide Antimonide (InAsPSb), and Gallium Indium Arsenide Antimonide Phosphide (GaInAsSbP).

(43) The discrete optical unit **243** can be a component that is separately formed from the supporting layer **242** and layers **210**, but can subsequently attached to a cavity **221** of the support layer **242** and/or layers **210**. The discrete optical unit **243** can include any type of material, such as silicon or glass, suitable for forming mirror(s) **209**, optic(s) **291**, or both. In some instances, the mirror **209** may redirect incident light to a top surface, such as the optics **291**.

(44) The discrete optical unit **243** can be located in a cavity **221** formed in the supporting layer **242** and/or layers **210**. The discrete optical unit **243** can be configured to receive light **241** and can redirect it towards the system interface **280**. In some examples, the discrete optical unit **243** can further include an anti-reflection coating **244** located on at least one facet, such as its receiving facet **245**.

(45) The mirror(s) **209** can be configured to redirect light. In some examples, the mirror(s) **209** can act as an outcoupler(s) configured to receive light from one plane (e.g., the plane located parallel to that of layers **210**) and redirect it to another plane (i.e., out of plane) (e.g., the plane located perpendicular to that of layers **210**). The mirror(s) **209** can be angled relative to the facet wall **222** of the cavity **221** and/or a receiving facet **245** of the discrete optical unit **243**. The receiving facet **245** can include an edge of the discrete optical unit **243** configured to receive light **241**. The angle

of the mirror(s) **209** can be any angle including, but not limited to, 45° and 54.7° . In some examples, the receiving surface (i.e., the surface where light **241** is incident on) of the mirror **209** can be configured with a profile shape that both turns and collimates (e.g., curved surface) light **241**.

(46) In some examples, at least a part of a side of the discrete optical unit **243** acts as the mirror **209**. The mirror may be formed such that light reflects at the interface of the discrete optical unit **243** and the gap **216** (or any material filling the space where the gap **216** is labeled in the figure). The mirror **209** may be formed in the gap **216**; for example, the mirror **209** may be formed at a surface of the discrete optical unit **243**. In such instances, the metal layer **208** may be excluded from the device **200**.

(47) In other instances, one or more layers may be deposited on a side (e.g., surface) of the discrete optical unit **243**, where the one or more layers can act as the mirror **209**. For example, the one or more layers may include one or more metal layers **208**. The light may be redirected at an interface of the side of the discrete optical unit **243** and a surface of the one or more metal layers **208**.

(48) In some examples, the one or more layers may form the mirror(s) **209** and can be made of the same material as the discrete optical unit **243**. The mirror may be such that light reflects at the interface of the one or more layers and the gap **216** (or any material filling the space where the gap **216** is labeled in the figure).

(49) In some examples, the discrete optical unit **243** can have certain dimensions (e.g., height and width as measured by the beginning and end of its angled wall) based on factors from other components. In some instances, the height of the mirror **209** can be configured such that a substantial part (e.g., all) of the beam from light **241** is incident on the angled wall of the mirror **209**. For example, the height of the mirror **209** can be based on the distance between where light **241** enters the cavity **221** (e.g., at the facet wall **222** of the cavity) and where light **241** is incident on the mirror **209**. In some instances, the location of where light **241** is incident on the mirror **209** can be determined based on the average of all locations of where light **241** is incident on the mirror **209**. For example, the distance from the facet wall **222** to the mirror **209** can be less for light **241** that is incident closer to the bottom of the cavity **221** than for light **241** that is incident closer to optics **291**. Since the distances from the facet wall **222** to the mirror **209** can vary, the average distance can be used to determine the height of the mirror **209**. The height of the mirror **209** can also be based on the height of where light **241** exits the facet wall **222** of the cavity (i.e., the termination point in the waveguide) to the bottom of the mirror **209**.

(50) In some examples, the height of the mirror **209** can be based on the divergence angles of light **241**. The divergence angles of light **241** may be affected by the material (if any) included between the facet wall **222** of the cavity **221** and the discrete optical unit **243**. For example, light **241** may have a greater divergence angle when only air is located between the facet wall **222** of the cavity **221** and the discrete optical unit **243** compared to other materials, such as a fill material (e.g., amorphous silicon).

(51) Optics **291** can be configured to redirect, collimate, and/or focus light towards the system interface **280**. The optics **291** can receive light from the mirror **209**. In some examples, the system interface **280** can be located between the measured sample volume (e.g., sample volume **120** illustrated in FIG. 1A) and the optics **291**. In some examples, optics **291** can include the same material as the discrete optical unit **243**. In some examples, the optics **291** can, at least in part, be formed from a portion of the discrete optical unit **243**. The optics **291** can include an anti-reflection coating **245** disposed on its top surface (i.e., surface closer to the system interface **280**).

(52) In some examples, the dimensions and placement of the optics **291** can be based on one or more factors such as the beam size and divergence. For example, the optics **291** can be located a certain separation distance from the mirror **209**. The separation distance can be selected such that light **241** can diverge by a certain amount (e.g., 10%) relative to mirror **209**. The amount of divergence can affect the size of the beam at the optics **291**. If the optics **291** is located too close to

the mirror **209**, then the beam size may be too small for the given application. In some instances, the separation distance can be determined using the average of the distances from the mirror **209**. (53) In some examples, the top surface (e.g., closer to the system interface **280**) of the discrete optical unit **243** can be located closer (i.e., protrude) to the system interface **280** than the top of layers **210**. The protrusion can allow light **241** to diverge from the mirror **209** to the optics **291** with reduced loss, as discussed earlier. The protrusion can be achieved in a number of ways such as by configuring the dimension (e.g., height) of the discrete optical unit **243** to be greater than the recess of the cavity **221** and/or by using a pedestal (discussed below).

(54) The device **200** can include one or more layers **210**. The layers **210** can include one or more conductive layers configured to route one or more signals to a light emitter (e.g., light emitter **107** illustrated in FIG. **1A**). For example, the layers **210** can be configured to route one or more signals from a controller (e.g., controller **140** illustrated in FIG. **1A**) to control the light emitter, which can emit light in response to the one or more signals. The layers **210** can also include one or more insulating layers. For example, the layers **210** can include multiple conductive layers electrically isolated by the insulating layer(s). In some instances, the layers **210** can include one or more encapsulation layers, passivation layers, planarizing layers, or the like.

(55) The layers **210** can also be used, at least in part, to form a waveguide for the light emitter. For example, the layer **210A** can include SiO₂ and can be used for insulating and/or encapsulating metal routing layers. The layer **210B** can be a silicon on insulator (SOI) layer, and the layer **210C** can be a buried oxide (BOX) layer.

(56) The device **200** can also include one or more adhesive layers **212** used to bond the discrete optical unit **243** to the supporting layer **242** and/or layers **210**. For example, the adhesive layer(s) **212** can be located between an attachment portion **214** of the discrete optical unit **243** and the supporting layer **242**. Optionally, the adhesive layer(s) **212** can be located between an attachment portion **214** of the discrete optical unit **243** and the layers **210** (as illustrated in FIG. **4D**). In such case, a gap (not shown) may be located between the discrete optical unit **243** and the bottom surface (e.g., perpendicular to the facet wall **222**) of the cavity **221**.

(57) As another option, the adhesive layer(s) **212** can be located between an attachment portion of the discrete optical unit **243** and a pedestal (as illustrated in FIG. **4E**). The terms “adhesive” or “adhesive layer” as used throughout this disclosure can include various different types of materials for attaching multiple components. Exemplary materials include, but are not limited to, glue, epoxy, and solder. Additionally, the terms can include an interface between the surfaces of multiple components due to a direct bond. In a direct bond, the surfaces can attach to each other using one or more phenomenon such as chemical bonding, Van der Waals force, etc.

(58) The discrete optical unit **243** can also include a gap **216** located between the metal layer **208** and the attachment portion **214**. The gap **216** can include air or any material such as glue. As discussed below, although the discrete optical unit **243** is illustrated as including a single mirror **209**, examples of the disclosure can include a discrete optical unit including multiple mirrors; some may be located in different cavities.

(59) Examples of the disclosure can include a fill material located between the facet wall **222** (and/or anti-reflection coating **223**) and the receiving facet **245** (and/or anti-reflection **244**). The properties of the fill material are discussed in more detail below.

(60) Operation of a Portion of an Exemplary Integrated Photonics Device

(61) FIG. **2B** illustrates an exemplary process showing the operation of a portion of an integrated photonics device according to examples of the disclosure. The integrated photonic device can be configured to emit light (e.g., towards a measured sample volume) out of its system interface (e.g., system interface **280** illustrated in FIG. **2A**). A light emitter (e.g., light emitter **107** illustrated in FIG. **1A**) can emit light (e.g., light **241** illustrated in FIG. **2A**) (step **252** of process **250**). The emitted light can propagate through a waveguide created by at least some of the layers (e.g., layers **210** illustrated in FIG. **2A**) (step **254** of process **250**). In some instances, some of the layers, at least

in part, form a cavity. The emitted light can be incident on a facet wall (e.g., facet wall **222** illustrated in FIG. **2A**) of a cavity, can enter (e.g., be received by) a portion of the cavity (e.g., cavity **221** illustrated in FIG. **2A**), and can diverge at certain angle(s) (step **256** of process **250**). The portion of the cavity that the emitted light enters can be the region located between the facet wall of the cavity and a facet of a discrete optical unit (e.g., discrete optical unit **243**). In some examples, the divergence angle(s) of the emitted light can be changed by a fill material located between the facet wall and a discrete optical unit. The emitted light can be incident on a facet of the discrete optical unit and can enter (e.g., be received by) it (step **258** of process **250**). A mirror (e.g., mirror **209** illustrated in FIG. **2A**) can redirect the light that has entered the discrete optical unit to the optics (e.g., optics **291** illustrated in FIG. **2A**) (step **260** of process **250**). The optics can direct (e.g., collimate and/or focus) the redirected light towards a system interface (e.g., system interface **280** illustrated in FIG. **2A**) (step **262** of process **250**).

(62) Overmold Layer

(63) In some examples, the integrated photonics device can include an overmold layer. FIG. **2C** illustrates a cross-sectional view of an exemplary portion of an integrated photonics device including an overmold layer according to examples of the disclosure. Device **201** can have corresponding similar structure and/or function as device **200**, including, but not limited to, components such as layers **210**, anti-reflection coating **223**, facet wall **222**, anti-reflection coating **244**, metal layer **208**, mirror **209**, gap **216**, adhesive layer **212**, attachment portion **214**, cavity **221**, and supporting layer **242** (discussed above).

(64) The device **201** may further include a discrete optical unit **247**, which can be similar to the discrete optical unit **243** of device **200** (illustrated in FIG. **2A**). In some examples, the discrete optical unit **247** may not include optics **291**. In some examples, the discrete optical unit **247** can have a height (e.g., distance from the bottom of cavity **221** to the top of the discrete optical unit **247**) that less than the height of the discrete optical unit **243**. The height of the discrete optical unit **247** may be such that the top of the discrete optical unit **247** protrudes outside (e.g., above) the cavity **221**.

(65) The device **201** can further include an overmold layer **233**. The overmold layer **233** can be located on top of layers **210**. In some examples, the overmold **233** can be deposited after a portion of the top of discrete optical unit **247** has been removed via, e.g., grinding. In some examples, the overmold **233** can be deposited after the discrete optical unit **247** is bonded to the supporting layer **242** and/or layers **210**. In some examples, a grinding step can be performed after the overmold layer **233** is deposited, which can lead to a reduced thickness of the overmold layer **233**.

(66) In some instances, the discrete optical unit **247** can be tested after the overmold layer is deposited. After the discrete optical unit **247** is tested, it can be removed (e.g., before or during the dies are diced). The discrete optical unit **247** may be removed if its performance does not meet one or more criteria. Alternatively, the discrete optical unit **247** can remain in the cavity after the dies are diced and used as an outcoupler. The discrete optical unit **247** can remain in the cavity if its performance meets one or more criteria.

(67) Multiple Discrete Optical Units

(68) Examples of the disclosure can include multiple discrete optical units. FIG. **3A** illustrates a top view of a portion of an integrated photonics device including multiple cavities according to examples of the disclosure. The supporting layer **342** can include a plurality of cavities, such as cavity **321A**, cavity **321B**, cavity **321C**, and cavity **321D**, some of which can be bonded to at least a portion of a discrete optical unit. In some examples, one or more cavities, such as cavity **321A**, can be used as attachment region(s) but not for receiving light. An attachment region can be a region for attaching the discrete optical unit to the supporting layer **342**.

(69) In some examples, the discrete optical unit can include multiple mirrors formed of the same substrate. FIG. **3B** illustrates a planar view of a portion of a discrete optical unit including multiple mirrors and attachment portions according to examples of the disclosure. The discrete optical unit

343 can include a plurality of mirrors **309**, a plurality of optics **391**, and a plurality of attachment portions **314**. In some examples, the discrete optical unit **343** can be formed from a silicon substrate. The discrete optical unit **343** can be bonded to the supporting layer at attachment portions **314** using an adhesive (including solder, direct bond, etc., as discussed above). When the discrete optical unit **343** is attached to the supporting layer (e.g., supporting layer **242** illustrated in FIG. 2A, supporting layer **342** illustrated in FIG. 3A), the attachment portions **314** can be located in one or more cavities of the supporting layer, outside of one or more cavities, or both. The optics **391** may also be located, at least in part, over (e.g., same line of sight) one or more cavities (e.g., cavity **221** illustrated in FIG. 2A, cavities **321** illustrated in FIG. 3A) of the supporting layer. In this manner, the discrete optical unit can provide a plurality of mirrors (e.g., outcouplers) for the device using a single substrate. Additionally, the plurality of outcouplers may be a two-dimensional array of outcouplers. In some examples, at least some of the plurality of outcouplers may be at locations inside from the edges of the device.

(70) Examples of the disclosure can also include other components, such as a plurality of detector arrays, included in the substrate that forms the discrete optical unit. FIG. 3C illustrates a top view of an exemplary discrete optical unit including detector arrays according to examples of the disclosure. The discrete optical unit **343** can include a plurality of detector arrays **330** formed and/or attached to the substrate. In some examples, the detector arrays **330** can be formed and/or attached before the discrete optical unit **343** is bonded to the supporting layer. In some examples, the substrate that forms the discrete optical unit **343** can include a plurality of metal traces (e.g., to route signals to the detector array). In this manner, a number of components can be included in the discrete optical unit: optics (e.g., on the other side of the discrete optical unit **343** from mirror **309**), outcouplers, detectors, traces (e.g., traces **331**), and the like.

(71) In some instances, an outcoupler can be associated with one of the detector arrays. For example, the mirror **309A** can be associated with the detector array **330A**, cavity **321A**, and corresponding traces **331A**. Light redirected by mirror **309A** to one region of the measured sample volume can return and be measured by the detector array **330A**. Similarly, the mirror **309C** can be associated with the detector array **330C**, cavity **321C**, and corresponding traces **331C**. Light redirected by mirror **309C** to another region of the measured sample volume can return and be measured by the detector array **330C**. Traces **331A** can transmit signals generated by detector **330A** and indicative of the return light measured by the detector array **330A**. Traces **331C** can transmit signals generated by detector **330C** and indicative of the return light measured by the detector array **330C**. A controller (e.g., controller **140** illustrated in FIG. 1A) can receive signals transmitted by traces **331A** and traces **331C** to determine one or more properties of the measured sample volume.

(72) The cavities **321**, mirrors **309**, attachment portions **314**, optics **391**, discrete optical unit **343**, and/or supporting layer **342** can have one or more properties and/or functions similar to the cavity **221**, mirror **209**, attachment portion **214**, optics **291**, discrete optical unit **243**, and supporting layer **242** discussed above in the context of FIG. 2A.

(73) Discrete Optical Units

(74) The integrated photonics device can also include a fill material. FIG. 4A illustrates a cross-sectional view of an exemplary portion of an integrated photonics device including a fill material according to examples of the disclosure. The fill material **432** can be deposited between the facet wall **422** of the cavity **421** and the receiving facet **445** of the discrete optical unit **443**. Without the fill material, this region would comprise air (e.g., as shown in FIG. 2A). The interface between air and the facet wall of the cavity and between air and the receiving facet of the discrete optical unit can cause a potential etalon. In some examples, the fill material **432** can include the same materials and/or layer structure as layers **410**. For example, the fill material **432** can be the same as layer **410A**, layer **410B**, and/or layer **410C**. In some instances, the fill material **432** can include the same material as the supporting layer **442**. In some examples, the fill material **432** can include amorphous silicon.

(75) In some examples, the discrete optical unit **443** can include one or more overhangs **433**. The figure illustrates an overhang **433** located to the left of optics **491**. Examples of the disclosure can include an overhang located to the right of the attachment portion **414** (e.g., as shown in FIG. 4F). The overhang **433** can be a portion of the discrete optical unit **443** located proximate to optics **491**. The overhang **433** can include the same materials as the mirror **409** and/or optics **491**. The overhang **433** can be included to capture portions of light **441** that has reflected off mirror **409**. For example, light **441A** may be lost without the overhang **433**. With the overhang **433**, light **441A** can be captured by other optics (not shown) and directed to the system interface **480**. In some examples, an integrated photonics device that includes the overhang **433** may not include a fill material **432**.

(76) In some examples, the discrete optical unit **443** can have an angled receiving facet **445** (e.g., angled relative to the bottom of the cavity **421**), as shown in FIG. 4B. The angled facet can facilitate the process of depositing the fill material **432**. In some instances, the fill material **432** may also have an angled edge. Additionally or alternatively, as shown in FIG. 4C, the facet wall **422** can be angled.

(77) In some instances, the device **400** can include one or more ledges **434**, as shown in FIG. 4D. The ledge **434** can be used to attach the discrete optical unit **443** to the supporting layer **442**, layers **410**, or both. The discrete optical unit **443** can attach to the ledge **434** via the attachment portion **414** using adhesive layer **412**. The ledge **434** can include any type of material such as those included in one or more of the layers **410**. For example, as shown in the figure, in forming the ledge **434**, the layer **410A** may not be deposited or may be removed, while layer **410B** and layer **410C** can be deposited or not removed. In some examples, the cavity **421** may have a smaller width than if the device did not include a ledge (e.g., as shown in FIG. 4A). In some examples, the ledge **434** may be a separate component that attaches to both the cavity **421** and the discrete optical unit **443** (not shown).

(78) The cavity may also include one or more pedestals, as shown in FIG. 4E. A pedestal **435** can be located at the bottom of the cavity **421**. The discrete optical unit **443** can be bonded to the pedestal **435** via its attachment portion **414** using an adhesive **412**. The pedestal **435** can be configured to provide the same location and/or plane for the bottom of the mirror **409**. In this manner, the recess of the cavity **421** can be etched and any changes in the depth of the recess can be compensated for with the pedestal **435**. In some examples, the height of the pedestal **435** can be based on the depth of the cavity **421**.

(79) FIG. 4F illustrates another example, where a side **436** of the attachment portion **414** can be angled wall relative to the bottom of the cavity **421**. The angled side **436** can help attach the discrete optical unit **443** to the supporting layer **442**. An adhesive **412** can be located between the angled side **436** and the supporting layer **442**, as shown. In some examples, the gap **416**, located between the mirror **409** and the attachment portion **414**, can be filled at least partially with adhesive.

(80) In some examples, the discrete optical unit may be a prism. FIG. 4G illustrates an exemplary discrete optical unit that includes a prism according to examples of the disclosure. The discrete optical unit **443** can have at least one side that forms a mirror **409**. The mirror **409** may, optionally, include a metal layer **408** deposited on the mirror **409**. In this manner, the mirror **409** can be the receiving facet of the discrete optical unit **443**. In some instances, the collimating optics can be located on a component (not shown) separate from the discrete optical unit **443**.

(81) The cavities **421**, mirrors **409**, attachment portions **414**, optics **491**, discrete optical unit **443**, supporting layer **442**, layers **410**, facet wall **422**, receiving facet **445**, and/or adhesive **412** can have one or more properties and/or functions similar to the cavity **221**, mirror **209**, attachment portion **214**, optics **291**, discrete optical unit **243**, supporting layer **242**, layers **210**, facet wall **222**, receiving facet **245**, and adhesive **212** discussed above in the context of FIG. 2A (and similarly, FIGS. 3A-3C). Additionally, the examples of the disclosure related to other components and

features such as the anti-reflection coating **223**, anti-reflection coating **244**, anti-reflection coating **245**, and gap **216** are applicable.

(82) Fabrication of a Portion of an Exemplary Integrated Photonics Device

(83) The discrete optical unit and the supporting layer can be formed in separate concurrent or serial processes. FIG. 5A illustrates an exemplary process for fabricating an exemplary discrete optical unit according to examples of the disclosure. The process can begin by forming the discrete optical unit (e.g., discrete optical unit **443** illustrated in FIGS. 4A-4G), which can include providing a substrate (step **552** of process **550**). One or more mirrors (e.g., mirror **409** illustrated in FIGS. 4A-4G) can be formed in a side (e.g., bottom side that faces the cavities when the discrete optical unit is attached to the supporting layer) of the discrete optical unit (step **554** of process **550**). In some examples, the discrete optical unit may include a prism, where the mirror can be a side of the prism (as shown in FIG. 4G). The mirror can be formed using any number of etching techniques such as gray scale etching, crystal plane etching, wet chemistry etching, and dry etching. Additionally, the mirror can be any angle (relative to a receiving facet of the discrete optical unit) including, but not limited to, 30°, 45°, 54.7°, 60°, etc. The angled wall of the mirror can be at any angle not coinciding with a crystallographic plane of the discrete optical unit. In some examples, a metal layer (e.g., metal layer **408** illustrated in FIGS. 4A-4G) may can be deposited on the mirror (step **556** of process **550**). In some examples, another angled side of the discrete optical unit can be formed. For example, side **436** illustrated in FIG. 4F can be formed before, after, or at the same time mirror **409** is formed.

(84) In some examples, a side of the discrete optical unit can be polished or etched to create an angled receiving facet (e.g., receiving facet **445** illustrated in FIG. 4B). In some examples, such as in devices where a fill material is not used, an anti-reflection coating (e.g., anti-reflection coating **244** illustrated in FIG. 2A) can be deposited on a receiving facet (e.g., receiving facet **445** illustrated in FIGS. 4A-4F) of the discrete optical unit (step **558** of process **550**).

(85) Optionally, one or more overhangs (e.g., overhang **433** illustrated in FIGS. 4A-4F) may be formed in one or more sides of the discrete optical unit (step **560** of process **550**). In some examples, optics (e.g., optics **491** illustrated in FIGS. 4A-4F) can be formed in another side (e.g., top side), opposite the first side, of the discrete optical unit (step **562** of process **550**). In some examples, the optics can be formed in another component, separate from the discrete optical unit.

(86) Additionally, in some examples, one or more process steps can include forming or attaching detector arrays (e.g., detector arrays **330** illustrated in FIG. 3C), traces (e.g., traces **331** illustrated in FIG. 3C), and/or other components in the discrete optical unit.

(87) FIG. 5B illustrates an exemplary process for forming the supporting layer and attaching the discrete optical unit to the supporting layer according to examples of the disclosure. A supporting layer (e.g., supporting layer **442** illustrated in FIGS. 4A-4G) and one or more (other) layers (e.g., layers **210** illustrated in FIG. 2A, layers **410** illustrated in FIGS. 4A-4G) can be provided (step **564** of process **551**). One or more cavities (e.g., cavity **421** illustrated in FIGS. 4A-4G) can be formed in the supporting layer and/or the other layers (step **566** of process **551**). The cavities can be formed using any type of etching technique such as dry etching or wet etching. In some instances, at least one wall of the cavity can be etched to form an angled wall (e.g., facet wall **422** illustrated in FIG. 4C). In some examples, the etching may be selective to form one or more ledges (e.g., ledge **434** illustrated in FIG. 4D).

(88) In some examples, one or more components can be provided and attached to the cavity. For example, a ledge (e.g., ledge **434** illustrated in FIG. 4D) and/or a pedestal (e.g., pedestal **435** illustrated in FIG. 4E) can be formed or bonded to the bottom of the cavity (e.g., cavity **421** illustrated in FIGS. 4D-4E). In some examples, such as in devices where a fill material is not used, an anti-reflection coating (e.g., anti-reflection coating **223** illustrated in FIG. 2A) can be deposited on at least one wall (e.g., facet wall **422** illustrated in FIGS. 4A-4G) of the cavity (step **568** of process **551**). A receiving facet of the discrete optical unit can be oriented towards a facet wall of a

cavity.

(89) Adhesive (e.g., adhesive **412** illustrated in FIGS. **4A-4G**) may optionally be dispensed at one or more attachment regions (step **570** of process **551**). The discrete optical unit(s) can be attached to the supporting layer at the attachment regions (step **572** of process **551**).

(90) In some examples, as discussed above in the context of FIG. **2C**, the process **551** can include a step (not shown) of removing a portion of the discrete optical unit (e.g., discrete optical unit **247**) to reduce its height. The process **551** may further include depositing an overmold layer (e.g., overmold layer **233** illustrated in FIG. **2C**) after the discrete optical unit (e.g., discrete optical unit **247** illustrated in FIG. **2C**) is attached in step **572**, and testing the discrete optical unit **247**.

Additionally, the process **551** may include removing the discrete optical unit **247** (if, e.g., the discrete optical unit **247** does not meet one or more criteria) or may include keeping the discrete optical unit **247** attached (if, e.g., the discrete optical unit meets one or more criteria).

(91) Optionally, a fill material (e.g., fill material **432** illustrated in FIGS. **4A-4B**) can be deposited in a portion of the cavity (e.g., cavity **421** illustrated in FIGS. **4A-4G**) of the supporting layer (step **574** of process **551**). In some examples, such as when the discrete optical unit is integrated into the supporting layer and other layers (e.g., layers **210** illustrated in FIG. **2A**, layers **410** illustrated in FIGS. **4A-4G**), the top surface of the discrete optical unit is polished such that it can be flush with the other layers.

(92) In some examples, the discrete optical unit can be bonded to the supporting layer by kinematic mounting. Negatively etched features can be formed in the bottom of the cavity, and protruding features can be formed in the discrete optical unit. The negatively etched features and protruding features can be kinematically mated. Additionally or alternatively, fiducials can be patterned on the discrete optical unit and the supporting layer for kinematic mounting. The fiducials can include features made from a metal layer or another material, for example.

(93) In other examples, the discrete optical unit can be bonded to the supporting layer using an active alignment process. The process can use an adhesive (e.g., solder, glue, direct bond, etc.), as discussed above, and one or more waveguides. The one or more waveguides can propagate light to the discrete optical unit. The light can be imaged externally or can reflect back to the detectors, which can be mounted on top of the discrete optical unit, for example. Light can be emitted by one or more light sources and/or a fiber coupled to the waveguide. The active alignment process can be performed either before or after multiple dies are separated (e.g., diced).

(94) In some examples, the collimating optics (e.g., optics **491** illustrated in FIGS. **4A-4F**) can be formed in the discrete optical unit after bonding to the supporting layer. An anti-reflection coating (e.g., anti-reflection coating **245** illustrated in FIG. **2A**) can be deposited on top of the collimating optics (e.g., optics **291** illustrated in FIG. **2A**), whether the collimating optics is formed before or after the discrete optical unit is bonded to the supporting layer.

(95) In instances where the discrete optical unit includes glass, a cure (e.g., ultraviolet cure) can be performed to set the adhesive (e.g., adhesive **412** illustrated in FIGS. **4A-4G**) used to adhere the discrete optical unit to the supporting layer, other layers, or both. In some examples, the discrete optical unit and the supporting layer can be bonded using a direct bonding. The contacting surfaces of the discrete optical unit and the supporting layer can be polished. Alternatively, the process of etching the cavity can lead to a surface in the supporting layer suitable for direct bonding.

(96) Multiple Waveguides and a Common Mirror

(97) Examples of the disclosure include an integrated photonics device having a plurality of waveguides. In some instances, the plurality of waveguides can share a common mirror. FIG. **6A** illustrates a top view of an exemplary integrated photonics device including a plurality of waveguides and a common mirror according to examples of the disclosure. The device **600** can include a cavity **621** and a plurality of waveguides **611**. The plurality of waveguides **611** can have termination points **612** located at the facet wall **622** of the cavity **621**. The cavity **621** can include the attachment region **613**, where the discrete optical units can be bonded to the supporting layer.

(98) In some examples, the waveguides may be located in a staggered (e.g., staircase) arrangement, as shown in FIG. 6B. The plurality of waveguides **611**, facet wall(s) **622** of the cavity **621**, and the termination points **612** can be arranged in a staggered manner. The plurality of waveguides can be associated with a plurality of mirrors **609**. In one example, the plurality of mirrors **609** can be separate and distinct discrete optical units, as shown in FIG. 6B.

(99) In another example, the plurality of mirrors **609** can be formed on a single discrete optical unit, as shown in FIG. 6C. The single discrete optical unit **643** having multiple mirrors arranged in a staggered manner. The spacing between the termination points **612** of the plurality of waveguides **611** and the mirrors can be the same. FIG. 6D illustrates a top view of an exemplary supporting layer **642** including a plurality of cavities **621** having staggered facet walls according to examples of the disclosure. In some examples, the discrete optical unit **643** can be attached to the supporting layer **642** in one or more regions where the cavities **621** are not located.

(100) In the above examples, the mirrors are not limited to having the same properties. Examples of the disclosure can include at least two mirrors having one or more of different angles, sizes, shapes, type of metal layer, and the like. Additionally, the locations of the mirrors are not limited to a one- or two-dimensional array of mirrors having equal spacing between mirrors. The mirrors can be placed anywhere along the device and in any manner.

(101) An integrated photonics device is disclosed. The integrated photonics device can include: a supporting layer including one or more cavities, at least one cavity including a facet wall; one or more light emitters that emit light towards the facet wall; a plurality of layers disposed on the supporting layer, wherein at least some of the plurality of layers are configured as one or more waveguides for the one or more light emitters; a discrete optical unit including: a substrate; one or more receiving facets that receive the emitted light, wherein the one or more receiving facets are formed from one or more sides of the substrate; one or more mirrors that redirect the emitted light received by the one or more receiving facets, wherein the one or more mirrors are formed from one or more sides of the substrate; and one or more attachment portions that bond the discrete optical unit to the supporting layer, wherein the one or more attachment portions are formed from one or more sides of the substrate; and an adhesive layer that bonds the one or more attachment portions of the discrete optical unit to the supporting layer, at least one of the plurality of layers, or both. Additionally or alternatively, in some examples, the integrated photonics device further comprises one or more antireflection coatings, the one or more anti-reflection coatings disposed on one or more of the facet wall and the one or more receiving facets. Additionally or alternatively, in some examples, the discrete optical unit further comprises at least one optic that receives the redirected light. Additionally or alternatively, in some examples, the at least one optics includes a same material as the discrete optical unit. Additionally or alternatively, in some examples, at least one of the one or more mirrors is angled relative to the facet wall and the one or more receiving facets. Additionally or alternatively, in some examples, a bottom of at least one of the one or more mirrors is located closer to a bottom of the at least one cavity than the plurality of layers. Additionally or alternatively, in some examples, the integrated photonics device further comprises a fill material located between the facet wall and at least one of the one or more receiving facets. Additionally or alternatively, in some examples, the fill material includes a same material as at least one of the plurality of layers. Additionally or alternatively, in some examples, the integrated photonics device further comprises: a system interface, wherein the plurality of layers is located closer to the system interface than the supporting layer, and further wherein a top of the discrete optical unit is located closer to the system interface than a top of the plurality of layers. Additionally or alternatively, in some examples, the plurality of layers includes a silicon on insulator (SOI) layer and a buried oxide (BOX) layer. Additionally or alternatively, in some examples, at least one of the one or more cavities is configured to attach to at least a portion of the discrete optical unit and does not receive the emitted light. Additionally or alternatively, in some examples, the integrated photonics device further comprises one or more detector arrays attached to the substrate of the discrete optical unit.

Additionally or alternatively, in some examples, the integrated photonics device further comprises a plurality of traces disposed on the substrate of the discrete optical unit. Additionally or alternatively, in some examples, termination points of the one or more waveguides are located in a same cavity, and one of the one or more mirrors is used to redirect the emitted light from the one or more waveguides. Additionally or alternatively, in some examples, at least two of the one or more mirrors have different angles.

(102) A method for emitting light out of a device is disclosed. The method comprises: emitting light by a light emitter, the light emitter included in the device; propagating the emitted light through a waveguide formed by at least some of a plurality of layers, wherein the waveguide and the plurality of layers are included in the device; receiving the emitted light at a facet wall of a cavity, the cavity included in a supporting layer of the device; receiving the emitted light at a receiving facet of a discrete optical unit, the discrete optical unit bonded to the supporting layer; redirecting the emitted light using a mirror of the discrete optical unit towards an optic; and directing the redirected light using the optic to a system interface of the device. Additionally or alternatively, in some examples, the method further comprises: changing one or more angles of divergence of the emitted light at the facet wall of the cavity using a fill material.

(103) A method for forming an integrated photonics device is disclosed. The method comprises: forming a discrete optical unit, wherein forming the discrete optical unit includes: providing a substrate; and forming one or more mirrors in one or more sides of the substrate; forming one or more cavities in a supporting layer, other layers, or both, wherein at least one of the one or more cavities include a facet wall; orienting a receiving facet of the discrete optical unit towards the facet wall of the at least one of the one or more cavities; and attaching the discrete optical unit to one or more of the supporting layer, the other layers, or both. Additionally or alternatively, in some examples, the method further comprises: depositing a fill material in the at least one of the one or more cavities between the receiving facet of the discrete optical unit and the facet wall of the at least one of the one or more cavities. Additionally or alternatively, in some examples, forming the discrete optical unit further comprises forming one or more overhangs in one or more sides of the substrate.

(104) Although the disclosed examples have been fully described with reference to the accompanying drawings, it is to be noted that various changes and modifications will become apparent to those skilled in the art. Such changes and modifications are to be understood as being included within the scope of the disclosed examples as defined by the appended claims.

Claims

1. An integrated photonics device including: a photonics substrate comprising: a supporting layer, and a plurality of additional layers disposed on the supporting layer and configured to define a waveguide, wherein: the photonics substrate defines a cavity extending through the plurality of additional layers and defining a facet wall in the plurality of additional layers; and the waveguide intersects the facet wall; and a discrete optical unit positioned at least partially in the cavity and connected to the supporting layer within the cavity, the discrete optical unit comprising: a receiving facet formed from a first surface of the discrete optical unit; a mirror formed at a second surface of the discrete optical unit; an attachment portion directly connected to a bottom surface of the cavity; and an overhang positioned above the plurality of additional layers, wherein: the receiving facet is positioned to receive light that exits the waveguide at the facet wall, and the mirror is positioned to redirect the light received by the receiving facet.
2. The integrated photonics device of claim 1, further comprising: one or more metal layers located at the second surface of the discrete optical unit, wherein the light is redirected at an interface at the second surface of the discrete optical unit and a surface of the one or more metal layers.
3. The integrated photonics device of claim 1, further comprising an antireflection coating disposed

on each of the facet wall and the receiving facet.

4. The integrated photonics device of claim 1, wherein the first surface, the second surface, and an exit surface of the discrete optical unit are integrated surfaces that form at least a portion of the discrete optical unit.

5. The integrated photonics device of claim 1, wherein the mirror is angled relative to the facet wall and the receiving facet.

6. The integrated photonics device of claim 1, wherein the mirror extends into the cavity past the plurality of additional layers.

7. The integrated photonics device of claim 1, further comprising a fill material located between the facet wall and the receiving facet.

8. The integrated photonics device of claim 7, wherein the fill material is made from the same material as at least one of the plurality of additional layers.

9. The integrated photonics device of claim 1, wherein the plurality of additional layers includes a silicon on insulator (SOI) layer and a buried oxide (BOX) layer.

10. The integrated photonics device of claim 1, comprising: a pedestal positioned in the cavity between the photonics substrate and the discrete optical unit, wherein the discrete optical unit is bonded to the pedestal.

11. The integrated photonics device of claim 1, wherein: the plurality of additional layers is configured to define a plurality of waveguides that includes the waveguide; the photonics substrate defines a plurality of cavities including the cavity, each of the plurality of cavities extending through the plurality of additional layers and defining a corresponding facet wall in the plurality of additional layers; and the discrete optical unit comprises a plurality of mirrors including the mirror; and each of the plurality of mirrors is positioned in a corresponding cavity of the plurality of cavities and is positioned to redirect light received by a corresponding waveguide of the plurality of waveguides.

12. The integrated photonics device of claim 11, wherein the discrete optical unit comprises a substrate, wherein the plurality of mirrors is formed from the substrate.

13. The integrated photonics device of claim 12, wherein the substrate comprises silicon.

14. The integrated photonics device of claim 12, wherein the discrete optical unit comprises a plurality of detector arrays positioned on the substrate.

15. The integrated photonics device of claim 14, wherein the discrete optical unit comprises a plurality of metal traces connected to the plurality of detector arrays.

16. The integrated photonics device of claim 1, wherein: the plurality of additional layers is configured to define a plurality of waveguides that includes the waveguide; the cavity defines a staggered plurality of facet walls in the plurality of additional layers that includes the facet wall; each waveguide of the plurality of waveguides intersects with a corresponding facet wall of the plurality of facet walls; the discrete optical unit comprises a staggered plurality of mirrors extending at least partially into the cavity and including the mirror; and each mirror of the plurality of mirrors is positioned to redirect light received by a corresponding waveguide of the plurality of waveguides.

17. The integrated photonics device of claim 16, wherein an attachment surface of the discrete optical unit is bonded to the supporting layer using an adhesive layer.

18. The integrated photonics device of claim 16, wherein the plurality of additional layers includes a silicon on insulator (SOI) layer and a buried oxide (BOX) layer.
